

Tech Products & Brand Perception: A Data Engineering Pipeline for Public Opinion and Sentiment Analysis

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Abstract—In the technology sector, understanding consumer perception is critical for brand management and product development. However, public opinion is dispersed across unstructured social media conversations and structured review platforms. This paper presents the design and implementation of an end-to-end data engineering and natural language processing (NLP) pipeline for sentiment analysis of technology products and brands. The proposed system integrates two complementary data sources: the X/Twitter API for real-time conversational data, and web scraping from GSMArena for structured user reviews. The pipeline follows the Medallion Architecture (Bronze, Silver, Gold) implemented with Apache Airflow for orchestration, PySpark for distributed data processing, and Plotly Dash for interactive visualization. A product marketing analyst is defined as the primary functional user. The system produces two interactive dashboards: a governance dashboard for data quality monitoring, and a storytelling dashboard for sentiment trend analysis. This paper describes the complete architecture, implementation methodology, and expected outcomes, demonstrating how modern data engineering tools can be applied to extract actionable insights from public opinion data.

I. INTRODUCTION

The proliferation of online platforms has transformed how consumers express opinions about technology products. From social media posts to specialized review websites, users generate vast amounts of unstructured and semi-structured textual data reflecting their experiences, preferences, and criticisms. For technology companies, this public discourse represents an invaluable resource for understanding brand perception, identifying product strengths and weaknesses, and monitoring competitive positioning [1].

However, extracting actionable insights from this data presents significant challenges. Public opinion is fragmented across multiple platforms with different formats, languages, and quality levels. Social media content is often informal, noisy, and context-dependent, while review platforms pro-

vide structured ratings alongside detailed textual feedback. Traditional manual analysis approaches are insufficient for processing data at scale, and many existing sentiment analysis tools fail to capture the nuanced language of technology discussions [2].

This work addresses these challenges by designing and implementing a complete data engineering pipeline for sentiment analysis of technology products and brands. The proposed system integrates two complementary data sources: the X/Twitter API (formerly Twitter) capturing real-time conversational sentiment, and web-scraped data from GSMArena, a specialized smartphone review platform. The pipeline follows the Medallion Architecture [3], with Bronze, Silver, and Gold layers orchestrating data ingestion, transformation, and aggregation.

Several prior works have explored sentiment analysis in the technology domain. [4] demonstrated that Twitter sentiment can predict box office revenues, establishing the value of social media analysis. [5] developed domain-specific sentiment lexicons for consumer electronics reviews. More recent approaches have leveraged deep learning for aspect-based sentiment analysis [6]. However, these works typically focus on either social media or reviews in isolation, without integrating both perspectives into a unified pipeline. Furthermore, few address the complete data engineering lifecycle from ingestion to visualization with production-grade tools.

The main contributions of this paper are: (1) a complete end-to-end data engineering pipeline for technology product sentiment analysis, (2) integration of complementary data sources (API and web scraping) to capture both conversational and evaluative sentiment, (3) implementation of the Medallion Architecture using Apache Airflow, PySpark, and Plotly Dash, and (4) a dual-dashboard approach providing both governance visibility and actionable business insights.

II. METHODS AND MATERIALS

A. System Architecture Overview

The proposed system implements a three-layer Medallion Architecture as illustrated in Figure 1. The Bronze layer handles raw data ingestion from both sources, preserving original JSON structures. The Silver layer performs data cleaning, text preprocessing, and sentiment classification, storing results in optimized Parquet format. The Gold layer aggregates data for analytical consumption, producing two summary datasets: governance KPIs and storytelling aggregations. Two independent Plotly Dash applications consume these Gold datasets, providing interfaces for technical and functional users respectively.

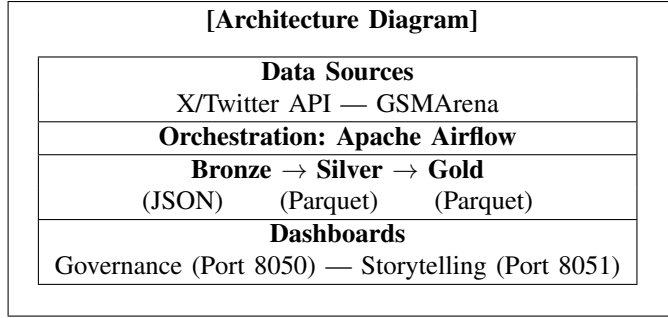


Fig. 1. High-level system architecture showing data flow from sources to dashboards.

B. Data Sources

1) *X/Twitter API*: The primary API-based source is the X/Twitter API accessed through the RapidAPI platform. This source provides real-time public conversations about technology products. Authentication uses API key-based methods, with the search endpoint <https://twitter-api45.p.rapidapi.com/search.php> as the primary extraction point. Key parameters include `query` (search keywords such as “iPhone”, “Samsung”, “Sony headphones”), `limit` (number of tweets per request), and `cursor` (pagination). The free tier allows approximately 50 requests per day, sufficient for daily ingestion of representative samples.

Extracted fields include: unique identifier, text content, creation timestamp, detected language, author information, engagement metrics (likes, retweets, replies), and URL. A validation sample of 20+ tweets confirmed data accessibility and relevance, with examples such as “I love my Sony xm4s” and “I love what #nothing is doing with the new 4a pro”.

2) *GSMArena Web Scraping*: The secondary source is GSMArena, a specialized smartphone review platform. The target pages are user review sections for specific device models. The HTML structure contains review text within `<div class="review-body">` elements, reviewer names in `<span class="reviewer-name">`, dates in `<div class="review-date">`, and numerical ratings in star rating elements.

Ethical scraping was verified by reviewing `robots.txt`, confirming that public review pages are permissible for crawling. To avoid server overload, requests will be spaced with delays and scheduled weekly rather than continuously. Extracted fields include reviewer, date, full review text, rating (1-10 scale), and product model. A validation sample of 40+ reviews for the Samsung Galaxy S23 demonstrated data accessibility.

C. Pipeline Implementation

1) *Bronze Layer: Raw Ingestion*: The Bronze layer is implemented as an Apache Airflow DAG using the TaskFlow API (`@dag` and `@task` decorators). The DAG runs on a schedule defined per source: daily for the X/Twitter API to capture dynamic social media content, and weekly for GSMArena scraping to respect server resources while maintaining current data.

Each ingestion run produces timestamped JSON files following the naming convention: `source_topic_YYYYMMDD_HHMMSS.json` (e.g., `x_smartphones_20260320_140530.json`). Files are stored in the `datalake_bronze/` directory, preserving original data structures for traceability and reproducibility. Error handling and logging are implemented at the task level to manage API rate limits and network failures.

2) *Silver Layer: Data Processing*: The Silver processing DAG is triggered by a FileSensor that monitors the `datalake_bronze/` directory for new JSON files. Upon detection, the preprocessing pipeline executes the following steps:

- 1) **Null handling**: Missing fields are either dropped, imputed (e.g., missing ratings filled with neutral values), or flagged for downstream analysis.
- 2) **Data type normalization**: Timestamps are converted to datetime objects, engagement metrics to integers, and categorical fields to appropriate types.
- 3) **Text preprocessing**: Lowercasing, punctuation removal, stop word removal, tokenization, and removal of URLs, mentions, and special characters.
- 4) **Sentiment classification**: Using TextBlob or a domain-adapted lexicon for polarity scoring (positive, negative, neutral).
- 5) **Deduplication**: Duplicate records are identified and removed based on content similarity and identifiers.
- 6) **Schema enforcement**: Output conforms to a defined Parquet schema with consistent field names and types.

Processed data is saved as Parquet files in `datalake_silver/` with the naming convention `topic_YYYYMMDD_HHMMSS.parquet`. Parquet was selected for columnar storage efficiency, compression, and compatibility with PySpark.

3) *Gold Layer: Aggregation and KPIs*: The Gold layer is implemented as a scheduled Airflow DAG using PySpark. The DAG reads all Parquet files from `datalake_silver/` using `spark.read.parquet("datalake_silver/")`, simulating a cloud storage folder read pattern. PySpark runs in local mode (`local[*]`) within the Docker environment.

Two summary datasets are produced:

Governance Summary: Computes KPIs including:

- Null rate per field (percentage of missing values)
- Total records ingested per source and time period
- Duplicate rate before and after deduplication
- Schema compliance rate
- Outlier rate per numeric field (using IQR and Z-score methods)
- Text length statistics (mean, median, min, max)
- Language distribution (if multilingual data present)
- Ingestion frequency compliance (actual vs. expected runs)

Storytelling Summary: Computes analytical aggregations for the functional user:

- Sentiment distribution (positive, negative, neutral counts and percentages)
- Sentiment trend over time (average sentiment score per day/week)
- Top keywords and n-grams with associated sentiment
- Source comparison (sentiment and volume by source)
- Volume trends (records per time period)
- Topic or category breakdown (sentiment per product feature or brand)

Both summaries are persisted as Parquet files in `datalake_gold/` with timestamps in the filename.

D. Docker and Orchestration Environment

The entire pipeline runs within Docker containers to ensure reproducibility and environment consistency. The Docker Compose configuration includes:

- **Airflow webserver:** Provides the user interface at `localhost:8080`
- **Airflow scheduler:** Executes DAG tasks according to schedules
- **Airflow triggerer:** Enables deferrable operators
- **PostgreSQL:** Metadata database for Airflow state
- **Volume mounts:** Local folders (`datalake_bronze/`, `datalake_silver/`, `datalake_gold/`, `airflow/dags/`) mounted for persistent data and live DAG reloading

Python dependencies include `requests`, `beautifulsoup4`, `pandas`, `polars`, `pyspark`, `pyarrow`, `textblob`, `dash`, `plotly`, and `dash-bootstrap-components`.

E. Dashboard Implementation

Two independent Plotly Dash applications are developed:

Governance Dashboard (`dashboard/governance_app.py`, port 8050):

- KPI summary cards (total records, null rates, duplicate rates)
- Null rate bar chart per field
- Volume over time chart (records per period per source)
- Outlier rate visualization per numeric field
- Data quality summary table with defined thresholds

Storytelling Dashboard (`dashboard/storytelling_app.py`, port 8051):

- Sentiment distribution donut chart
- Sentiment trend line chart over time
- Top keywords bar chart with sentiment coloring
- Source comparison chart
- Volume activity chart
- Narrative summary card with key insights

Both dashboards load data exclusively from Gold Parquet files using `pd.read_parquet()`, ensuring the Gold layer serves as the single source of truth for presentation.

F. Functional User Definition

The primary functional user is a product marketing analyst in the technology sector. This user is responsible for monitoring public perception of technological products and identifying trends in user opinions. Technical level is low to moderate; the user does not interact with raw data or code but relies on intuitive dashboards. Information needs include overall sentiment, changes over time, and most frequently discussed features. The user benefits from data-driven insights for marketing strategies, product improvements, and competitive analysis.

G. User Stories

Six user stories guide dashboard design:

- 1) As a product marketing analyst, I want to view overall sentiment trends per brand over time so that I can identify whether public perception is improving or declining after marketing campaigns.
- 2) As a product marketing analyst, I want to compare sentiment across competing brands (Apple, Samsung, Google) so that I can benchmark our brand's performance against competitors.
- 3) As a product marketing analyst, I want to see which product features (battery life, camera, display) are most frequently mentioned in positive versus negative reviews so that I can understand what resonates with users.
- 4) As a product marketing analyst, I want to detect sudden spikes or drops in sentiment related to product launches so that I can react quickly to emerging issues.
- 5) As a product marketing analyst, I want to filter sentiment data by source (X/Twitter vs. GSMarena) so that I can distinguish between social media buzz and structured reviews.
- 6) As a product marketing analyst, I want to view a dashboard that updates daily with fresh sentiment insights so that I can rely on current data for weekly reports.

III. EXPECTED RESULTS AND DISCUSSION

A. Expected Outcomes

Upon full implementation, the system is expected to produce:

- 1) A fully functional data pipeline orchestrated by Apache Airflow with three DAGs (Bronze ingestion, Silver processing triggered by FileSensor, and Gold aggregation).

- 2) A collection of timestamped JSON files in the Bronze layer representing raw data from both sources across multiple ingestion runs.
- 3) Parquet files in the Silver layer containing cleaned, preprocessed, and sentiment-labeled data.
- 4) Two Gold layer Parquet files: one with governance KPIs and one with storytelling aggregations.
- 5) Two interactive Plotly Dash dashboards accessible at `localhost:8050` and `localhost:8051`, displaying real data from the Gold layer.
- 6) A reproducible Docker environment where the entire pipeline can be launched with `docker-compose up` from any clean machine.

B. Anticipated Challenges

Several challenges are anticipated based on the data samples and pipeline design:

Data Quality: The collected samples reveal multiple languages (English, French, Indonesian), informal language, emojis, and user mentions. The preprocessing pipeline must handle these effectively, potentially incorporating language detection and translation for multilingual content.

Sentiment Accuracy: Informal technology discourse often includes sarcasm, technical jargon, and comparative language (e.g., “iPhone is good but Samsung is better”). The sentiment classification approach using TextBlob may require domain adaptation or lexicon augmentation for technology-specific terms.

API Rate Limits: The X/Twitter API free tier permits approximately 50 requests per day. This may limit sample size, requiring careful selection of search keywords and potential scheduling optimizations.

PySpark Performance: Reading all Silver Parquet files with PySpark in local mode must handle growing data volumes over the semester. The DAG design should account for potential memory constraints and consider incremental processing strategies.

C. Evaluation Metrics

System success will be evaluated using:

- **Functional completeness:** All DAGs execute without errors, producing expected outputs.
- **Data quality:** Governance KPIs demonstrate acceptable null rates ($\leq 5\%$) and duplicate rates ($\leq 2\%$).
- **Sentiment accuracy:** Manual validation of a sample of 100 records against predicted sentiment, targeting $\geq 80\%$ accuracy.
- **Reproducibility:** Successful deployment from a fresh repository clone following README instructions.
- **User satisfaction:** The functional user can answer the six user stories using the storytelling dashboard without technical assistance.

D. Integration of API and Web Scraping Perspectives

A key contribution of this work is the integration of two complementary sentiment perspectives. The X/Twitter

data captures real-time, conversational sentiment—immediate reactions to product launches, announcements, and trends. GSMArena reviews provide structured, evaluation-oriented sentiment from users with hands-on experience, including numerical ratings that can serve as ground truth.

By combining these sources, the pipeline enables analyses that neither source alone could support. For example, a product launch might generate immediate positive buzz on X/Twitter but later reveal issues in detailed reviews. The dual-dashboard approach maintains separate governance and storytelling interfaces while sharing the same underlying data integration.

IV. CONCLUSIONS

This paper presented the design of an end-to-end data engineering pipeline for sentiment analysis of technology products and brands. The system integrates two complementary data sources—X/Twitter API and GSMArena web scraping—through a three-layer Medallion Architecture implemented with Apache Airflow, PySpark, and Plotly Dash. The pipeline addresses the complete data lifecycle from raw ingestion to interactive visualization, producing both governance dashboards for data quality monitoring and storytelling dashboards for business insights.

The work is grounded in real data validation, with sample collections from both sources confirming accessibility and NLP feasibility. The functional user definition and user stories provide clear guidance for dashboard design, ensuring the final system delivers actionable insights to product marketing analysts.

Current limitations include dependency on API rate limits, the challenge of multilingual and informal language processing, and the use of basic sentiment classification techniques. Future work will explore aspect-based sentiment analysis to capture opinions about specific product features (battery, camera, display), integration of additional data sources such as YouTube comments or Reddit discussions, and deployment to cloud infrastructure for scalability.

The next phases of implementation will focus on completing the Airflow DAGs, implementing the PySpark transformations, and developing the Plotly Dash applications. The complete system will be open-sourced on GitHub, providing a reproducible template for similar sentiment analysis projects in other domains.

V. ACKNOWLEDGMENT

The authors acknowledge the support of the Computer Engineering Program at Universidad Distrital Francisco José de Caldas, and the guidance of Professor Eng. Carlos Andrés Sierra throughout this project.

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